The structure of Zhu's algebras for triplet vertex operator (super)algebras

Dražen Adamović

University of Zagreb, Croatia

Beijing, June 16, 2011

In this talk we shall study the structure of Zhu's algebras for certain vertex operator (super)algebra.

- We determine Zhu's algebra for :

- the triplet vertex algebra $\mathcal{W}(p)$
- the $N = 1$ triplet vertex superalgebra $\mathcal{SW}(m)$;
- the $c = 0$ triplet vertex algebra $\mathcal{W}_{2,3}$.

- The existence of logarithmic representations is obtained from the structure of Zhu's algebra.
- We determine the Poisson algebras $\mathcal{P}(\mathcal{W}(p))$ and $\mathcal{P}(\mathcal{SW}(m))$.

This talk is based on a joint work with Antun Milas.

In this talk we shall study the structure of Zhu's algebras for certain vertex operator (super)algebra.

- We determine Zhu's algebra for :

- the triplet vertex algebra $\mathcal{W}(p)$
- the $N = 1$ triplet vertex superalgebra $\mathcal{SW}(m)$;
- the $c = 0$ triplet vertex algebra $\mathcal{W}_{2,3}$.

- The existence of logarithmic representations is obtained from the structure of Zhu's algebra.
- We determine the Poisson algebras $\mathcal{P}(\mathcal{W}(p))$ and $\mathcal{P}(\mathcal{SW}(m))$.

This talk is based on a joint work with Antun Milas.

In this talk we shall study the structure of Zhu's algebras for certain vertex operator (super)algebra.

- We determine Zhu's algebra for :

- the triplet vertex algebra $\mathcal{W}(p)$
- the $N = 1$ triplet vertex superalgebra $\mathcal{SW}(m)$;
- the $c = 0$ triplet vertex algebra $\mathcal{W}_{2,3}$.

- The existence of logarithmic representations is obtained from the structure of Zhu's algebra.
- We determine the Poisson algebras $\mathcal{P}(\mathcal{W}(p))$ and $\mathcal{P}(\mathcal{SW}(m))$.

This talk is based on a joint work with Antun Milas.

In this talk we shall study the structure of Zhu's algebras for certain vertex operator (super)algebra.

- We determine Zhu's algebra for :

- the triplet vertex algebra $\mathcal{W}(p)$
- the $N = 1$ triplet vertex superalgebra $\mathcal{SW}(m)$;
- the $c = 0$ triplet vertex algebra $\mathcal{W}_{2,3}$.

- The existence of logarithmic representations is obtained from the structure of Zhu's algebra.
- We determine the Poisson algebras $\mathcal{P}(\mathcal{W}(p))$ and $\mathcal{P}(\mathcal{SW}(m))$.

This talk is based on a joint work with Antun Milas.

D. A. and A. M., On the triplet vertex algebra $\mathcal{W}(p)$, Advances in Mathematics (2008)

D. A. and A. M., The $N = 1$ triplet vertex operator superalgebras, *Communications in Mathematical Physics* (2009)

D. A and A. M., Lattice construction of logarithmic modules for certain vertex algebras, Selecta Mathematica, New Series 15 (2009)

D.A. and A.M., On W-Algebras Associated to (2, p) Minimal Models and Their Representations, IMRN (2010)

D. A. and A. M., The structure of Zhu's algebras for certain W-algebras, Advances in Mathematics (2011)

Related works on the triplet vertex algebras:

H. Kausch (1991)

Kausch-Gaberdiel (1996)

M. Flohr (1996)

Fjelsted-Fuchs-Hwang-Semikhatov- Tipunin (2003)

Abe (2005)

Flohr-Carqueville(2005)

Feigin-Gainutdinov-Semikhatov-Tipunin (2005; 2006)

J.Fuchs (2006)

A. Semikhatov (2007; 2008; 2009; 2010)

K. Nagatomo, A. Tsuchiya (2009)

Y.Z. Huang (2009)

M. Gaberdiel, I. Runkel, S. Wood (2009; 2010)

Vertex operator superalgebras

Let $(V, Y, \mathbf{1}, \omega)$ be a vertex operator superalgebra,

$V = V_{\bar{0}} \oplus V_{\bar{1}}$ is a $\mathbb{Z}_2$–graded vector space,

$\mathbf{1} \in V_{\bar{0}}$ is the vacuum of V,

ω is the conformal vector,

and Y is a vertex operator map

$$Y(\cdot, z) : V \to (\text{End } V)[[z, z^{-1}]];$$
$$a \mapsto Y(a, z) = \sum_{n \in \mathbb{Z}} a_n z^{-n-1} \in (\text{End } V)[[z, z^{-1}]].$$

If $V = V_{\bar{0}}$, V is vertex operator algebra.

Zhu's algebra $A(V)$

We define two bilinear maps $* : V \times V \to V$, $\circ : V \times V \to V$ as follows.

For homogeneous $a, b \in V$ let

$$a * b = \begin{cases} \mathrm{Res}_z Y(a,z)\frac{(1+z)^{\deg(a)}}{z} b & \text{if } a, b \in V_{\bar{0}} \\ 0 & \text{if } a \text{ or } b \in V_{\bar{1}} \end{cases}$$

$$a \circ b = \begin{cases} \mathrm{Res}_z Y(a,z)\frac{(1+z)^{\deg(a)}}{z^2} b & \text{if } a \in V_{\bar{0}} \\ \mathrm{Res}_z Y(a,z)\frac{(1+z)^{\deg(a)-\frac{1}{2}}}{z} b & \text{if } a \in V_{\bar{1}} \end{cases}$$

$O(V) = \mathrm{span}_{\mathbb{C}}\{a \circ b \mid a, b \in V\}$.

$A(V) = V/O(V)$.

Zhu's algebra

$(A(V), *)$ is associative, unital algebra, called Zhu's algebra of V

There is one-to-one correspondence between irreducible $A(V)$–modules and irreducible $\frac{1}{2}\mathbb{Z}_{\geq 0}$–graded V–modules.

Moreover, if U is any $A(V)$–module. There is $\frac{1}{2}\mathbb{Z}_{\geq 0}$–graded V–module $L(U)$ such that the top component $L(U)(0) \cong U$.

V is called rational, if V has finitely many irreducible modules and the category of modules is semisimple.

If V is rational, then $A(V)$ is semisimple.

Twisted Zhu's algebra $A_\sigma(V)$

Let V be a vertex operator superalgebra and σ the canonical $\mathbb{Z}_2$–automorphism

Let $O_\sigma(V)$ be the vector space spanned by elements of the form

$$\mathrm{Res}_z \frac{(1+z)^{\deg a}}{z^2} Y(a,z)b \quad (a \textit{ homogeneous}).$$

The vector space $A_\sigma(V) = V/O_\sigma(V)$ is equipped with the structure of an associative algebra via:

$$a * b = \mathrm{Res}_z \frac{(1+z)^{\deg a}}{z} Y(a,z)b.$$

C_2–cofinite vertex operator superalgebras

Define

$$C_2(V) = \mathrm{span}_{\mathbb{C}}\{a_{-2}b \mid a, b \in V\}.$$

Let $\mathcal{P}(V) = V/C_2(V)$.

For $a \in V$ we set $\overline{a} = a + C_2(V)$.
$\mathcal{P}(V)$ is an algebra with multiplication:

$$\overline{a} \cdot \overline{b} = \overline{a_{-1}b}, \qquad (a, b \in V).$$

V is called C_2–cofinite if $\dim \mathcal{P}(V) < \infty$.

C_2–cofinite vertex operator algebras

Theorem (Zhu)

If V is C_2–cofinite, then Zhu's algebra $A(V)$ is finite dimensional, and therefore V has finitely many irreducible modules.

Theorem (Dong, Jiang)

If vertex operator algebra V is C_2–cofinite, then V is rational if and only if $A(V)$ is semisimple.

Zhu's algebra and C_2–algebra

Let V be a C_2–cofinite vertex operator algebra. M. Gaberdiel and T. Gannon (2009) initiated a relationship between $A(V)$ and $\mathcal{P}(V)$.

Question: When $\dim A(V) = \dim \mathcal{P}(V)$?

For a large family of rational VOAs of affine type, the equality of dimensions holds: B. Feigin, E. Feigin, P. Littelman (2010-2011).

We study this question for C_2–cofinite, irrational vertex operator algebra.

If V is a vertex operator superalgebra, one can expect that $\dim \mathcal{P}(V) = \dim A_\sigma(V)$.

Virasoro vertex operator algebras

Let $Vir = \oplus_{n\in\mathbb{Z}}\mathbb{C}L(n) \oplus \mathbb{C}C$ be a Virasoro algebra, i.e.,
$[L(m), L(n)] = (m-n)L(m+n) + \frac{m^3-m}{12}\delta_{m+n,0}C$
C is central element.

Let $V(c,0)$ be the universal Virasoro VOA with central charge c.

Let $L(c,0)$ be the simple Virasoro VOA with central charge c.

Let $L(c,h)$ be irreducible highest weight module of Vir (and of the VOA $V(c,0)$) with central charge c and highest weight h.

A homomorphism of Zhu's algebras

Assume that V is a vertex operator superalgebra which admits the (horizontal) $\mathbb{Z}$–gradation:

$$V = \bigoplus_{\ell \in \mathbb{Z}} V_\ell, \quad V_{\ell_1} \cdot V_{\ell_2} \subset V_{\ell_1+\ell_2},$$

where

$$V_k \cdot V_\ell := \mathrm{span}_{\mathbb{C}}\{u_n v : u \in V_k, v \in V_\ell, n \in \mathbb{Z}\}.$$

In addition, assume there is $G \in \mathrm{End}(V)$ such that:

$$G \quad \text{is a derivation on V}, \tag{1}$$

$$G(V_\ell) \subset V_{\ell+1}, \tag{2}$$

$$\omega \in V_0, \quad [G, Y(\omega, z)] = 0, \tag{3}$$

$$G|V_\ell \quad \text{is injective for} \quad \ell < 0, \tag{4}$$

$$G|V_\ell \quad \text{is surjective for} \quad \ell \geq 0. \tag{5}$$

A homomorphism of Zhu's algebras

Clearly, V_0 is a subalgebra of V.

We shall consider Zhu's algebras

$$A(V_0) = V_0/O(V_0), \quad A(V) = V/O(V),$$

and natural homomorphism:

$$\begin{aligned} \Phi : \quad & A(V_0) \to A(V) \\ & v + O(V_0) \mapsto v + O(V), \quad v \in V_0. \end{aligned}$$

Let

$$A(V)_0 = \mathrm{Im}(\Phi) \cong A(V_0)/\mathrm{Ker}(\Phi).$$

A homomorphism of Zhu's algebras

We shall consider vertex operator superalgebras such that

$$A(V) = A(V)_{-1} \oplus A(V)_0 \oplus A(V)_1;$$

where $A(V)_{\pm 1}$ are $A(V)_0$–modules.

Lattice vertex algebra V_L

Let $V_L = M(1) \otimes \mathbb{C}[L]$ be the vertex algebra associated to the lattice

$$L = \mathbb{Z}\alpha, \quad \langle \alpha, \alpha \rangle = 2pp', \ p, p' \in \mathbf{Z}_{>0}, \ p \geq 2.$$

where $M(1)$ is the associated the Heisenberg vertex algebra, and $\mathbb{C}[L]$ is the group algebra with generator e^{α}.

Let Y be the associated vertex operator map.

Then V_L has the structure of a vertex operator algebra of rank $c_{p,p'}$ with conformal vector

$$\omega = \frac{1}{4pp'}\alpha(-1)^2 + \frac{p-p'}{2pp'}\alpha(-2) \in M(1).$$

The vertex operator algebras $\mathcal{W}(p)$ and $\mathcal{W}_{p,p'}$

The associated screening operators are

$$Q = \mathrm{Res}_z Y(e^{\alpha/p'}, z), \quad \widetilde{Q} = \mathrm{Res}_z Y(e^{-\alpha/p}, z).$$

The following vertex algebra was defined by B. L. Feigin, A. M. Gainutdinov,A.M. Semikhatov, I. Yu Tipunin (2006):

$$\boldsymbol{\mathcal{W}}_{p,p'} = \mathrm{Ker}_{V_L} Q \cap \mathrm{Ker}_{V_L} \widetilde{Q}.$$

We also have the following $\mathcal{W}$–algebra:

$$\overline{M(1)}_{p,p'} = \mathrm{Ker}_{M(1)} Q \cap \mathrm{Ker}_{M(1)} \widetilde{Q}.$$

$\boldsymbol{\mathcal{W}}_{p,p'}$ and $\overline{M(1)}_{p,p'}$ are (logarithmic) extensions of the Virasoro vertex operator algebra $V(c_{p,p'}, 0)$.

Assume that $p' = 1$. Define

$$\mathcal{W}(p) = \mathrm{Ker}_{V_L} \widetilde{Q}, \quad \overline{M(1)}_p = \mathrm{Ker}_{M(1)} \widetilde{Q}.$$

The triplet vertex algebra $\mathcal{W}(p)$

Assume that $p' = 1$. Define

$$\mathcal{W}(p) = \mathrm{Ker}_{V_L} \widetilde{Q}.$$

$\mathcal{W}(p)$ is a (logarithmic) extension of $L(c_{p,1}, 0)$.
$\mathcal{W}(p)$ is $\mathcal{W}$–algebra of type $\mathcal{W}(2, 2p-1, 2p-1, 2p-1)$.
Generators:

$$\omega, \ F = e^{-\alpha}, \ H = QF, \ E = Q^2 F.$$

$\overline{M(1)}_p$ is $\mathcal{W}$–algebra of type $\mathcal{W}(2, 2p-1)$ generated by ω and H.

The triplet vertex algebra $\mathcal{W}(p)$

Assume that $p' = 1$. Define

$$\mathcal{W}(p) = \mathrm{Ker}_{V_L} \widetilde{Q}.$$

$\mathcal{W}(p)$ is a (logarithmic) extension of $L(c_{p,1}, 0)$.
$\mathcal{W}(p)$ is $\mathcal{W}$–algebra of type $\mathcal{W}(2, 2p-1, 2p-1, 2p-1)$.
Generators:

$$\omega,\ F = e^{-\alpha},\ H = QF,\ E = Q^2 F.$$

$\overline{M(1)}_p$ is $\mathcal{W}$–algebra of type $\mathcal{W}(2, 2p-1)$ generated by ω and H.

The triplet vertex algebra $\mathcal{W}(p)$

$\mathcal{W}(p)$ is $\mathbb{Z}$–graded

$$\mathcal{W}(p) = \bigoplus_{\ell \in \mathbb{Z}} \mathcal{W}(p)_\ell,$$

where

$$\mathcal{W}(p)_{-\ell} = \overline{M(1)}_p . e^{-\ell\alpha}, \quad \mathcal{W}(p)_\ell = \overline{M(1)}_p . Q^{2\ell} e^{-\ell\alpha}$$

$G = Q$ is derivation.

Zhu's algebra $\overline{M(1)}_p$

Let $h_{r,s} = \frac{(sp-r)^2-(p-1)^2}{4p}$.

Theorem

Zhu's associative algebra $A(\overline{M(1)}_p)$ is isomorphic to the commutative algebra $\mathbb{C}[x,y]/\langle P(x,y)\rangle$, where $\langle P(x,y)\rangle$ is the principal ideal generated by

$$P(x,y) = y^2 - C_p(x - h_{p,1})\prod_{i=1}^{p-1}(x - h_{i,1})^2 \qquad (C_p \neq 0).$$

(Here x and y correspond to $[\omega]$ and $[H]$).

Zhu's algebra $A(\mathcal{W}(p))$

Consider homomorphism $\Phi : A(\overline{M(1)}_p) \to A(\mathcal{W}(p))$.

We have:

$$\mathrm{Ker}(\Phi) = A(\overline{M(1)}_p).(p([\omega]) * [H])$$

where $p(x) = \prod_{i=2p}^{3p-1}(x - h_{i,1})$.

Proposition.

$$A(\mathcal{W}(p)) = A(\mathcal{W}(p))_{-1} \oplus A(\mathcal{W}(p))_0 \oplus A(\mathcal{W}(p))_1,$$

where

$$A(\mathcal{W}(p))_{-1} = A(\mathcal{W}(p))_0.[F], \quad A(\mathcal{W}(p))_1 = A(\mathcal{W}(p))_0.[E],$$

$$\dim A(\mathcal{W}(p))_{\pm 1} = p, \quad \dim A(\mathcal{W}(p))_0 = 4p - 1.$$

The triplet vertex algebra $\mathcal{W}(p)$

Theorem (D.A, A.M)

.

(i) *$\mathcal{W}(p)$ has exactly $2p$ irreducible modules:*

$$\Lambda(1), \ldots, \Lambda(p); \Pi(1), \ldots, \Pi(p).$$

(ii) *$\mathcal{W}(p)$ is C_2–cofinite.*

(iii)

$$\dim A(\mathcal{W}(p)) = \dim \mathcal{P}(\mathcal{W}(p)) = 6p - 1.$$

The triplet vertex algebra $\mathcal{W}(p)$

Theorem (D.A, A.M)

.

(i) *$\mathcal{W}(p)$ has exactly $2p$ irreducible modules:*

$$\Lambda(1), \ldots, \Lambda(p); \Pi(1), \ldots, \Pi(p).$$

(ii) *$\mathcal{W}(p)$ is C_2–cofinite.*

(iii)

$$\dim A(\mathcal{W}(p)) = \dim \mathcal{P}(\mathcal{W}(p)) = 6p - 1.$$

The triplet vertex algebra $\mathcal{W}(p)$

Theorem (D.A, A.M)

.

(i) *$\mathcal{W}(p)$ has exactly $2p$ irreducible modules:*

$$\Lambda(1), \ldots, \Lambda(p); \Pi(1), \ldots, \Pi(p).$$

(ii) *$\mathcal{W}(p)$ is C_2–cofinite.*

(iii)

$$\dim A(\mathcal{W}(p)) = \dim \mathcal{P}(\mathcal{W}(p)) = 6p - 1.$$

The triplet vertex algebra $\mathcal{W}(p)$

Theorem (D.A, A.M)

.

(i) $\mathcal{W}(p)$ *has exactly* $2p$ *irreducible modules:*

$$\Lambda(1), \ldots, \Lambda(p); \Pi(1), \ldots, \Pi(p).$$

(ii) $\mathcal{W}(p)$ *is* C_2*–cofinite.*

(iii)

$$\dim A(\mathcal{W}(p)) = \dim \mathcal{P}(\mathcal{W}(p)) = 6p - 1.$$

Zhu's algebra $A(\mathcal{W}(p))$

Generators of $A(\mathcal{W}(p))$:

$$[\omega], [H], [E], [F].$$

Theorem

Zhu's algebra $A(\mathcal{W}(p))$ decomposes as a direct sum:

$$A(\mathcal{W}(p)) = \bigoplus_{i=2p}^{3p-1} \mathbb{M}_{h_{i,1}} \oplus \bigoplus_{i=1}^{p-1} \mathbb{I}_{h_{i,1}} \oplus \mathbb{C}_{h_{p,1}},$$

where

$\mathbb{M}_{h_{i,1}}$ *ideal isomorphic to matrix algebra* $M_2(\mathbb{C})$,

$\mathbb{I}_{h_{i,1}}$ *is 2–dimensional ideal,*

$\mathbb{C}_{h_{p,1}}$ *is 1-dimensional ideal.*

Structure of $\mathcal{P}(\mathcal{W}(p))$

Generators of $\mathcal{P}(\mathcal{W}(p))$:

$$\bar{\omega}, \bar{H}, \bar{E}, \bar{F},$$

Relations:

$$\begin{aligned}
&\bar{E}^2 = \bar{F}^2 = \bar{H}\bar{F} = \bar{H}\bar{E} = 0, \\
&\bar{H}^2 = -\bar{E}\bar{F} = \nu\bar{\omega}^{2p-1} \quad (\nu \neq 0), \\
&\bar{\omega}^p\bar{H} = \bar{\omega}^p\bar{E} = \bar{\omega}^p\bar{F} = 0.
\end{aligned}$$

Logarithmic modules

Corollary

For every $1 \leq i \leq p-1$, there exits the logarithmic, self-dual, $\mathbb{Z}_{\geq 0}$–graded $\mathcal{W}(p)$–module $\mathcal{P}_i^+$ such that the top component $\mathcal{P}_i^+(0)$ is two-dimensional and $L(0)$ acts on $\mathcal{P}_i^+(0)$ as

$$\begin{pmatrix} h_{i,1} & 1 \\ 0 & h_{i,1} \end{pmatrix}.$$

Remark.

$\mathcal{W}(p)$ also has $(p-1)$–logarithmic modules $\mathcal{P}_i^-$ which can not be detected by $A(\mathcal{W}(p))$. Construction:

$$\mathcal{P}_i^- := \mathcal{P}_{p-i}^+ \boxtimes \Pi(p-1).$$

Triplet vertex algebra $\mathcal{W}_{p,2}$

The triplet vertex algebra $\mathcal{W}_{p,2}$ can be realized as a subalgebra of V_L generated by ω and primary vectors

$$F = Qe^{-3\alpha/2}, \quad H = GF, \quad E = G^2F,$$

where G is (new) screening operator defined by

$$G = \sum_{i=1}^{\infty} \frac{1}{i} e_{-i}^{\alpha/2} e_i^{\alpha/2}.$$

Moreover, $\overline{M(1)}_{p,2}$ is generated by ω and H.

Zhu's algebra $\overline{M(1)}_{p,2}$

$$h_{r,s} = \frac{(pr-2s)^2-(p-2)^2}{8p}.$$

Theorem

Zhu's algebra of the singlet vertex algebra $A(\overline{M(1)}_{p,2})$ is isomorphic to the commutative associative algebra

$$A(\overline{M(1)}_{p,2}) \cong \frac{\mathbb{C}[x,y]}{\langle P(x,y)\rangle}$$

where

$$P(x,y) = y^2 - \widetilde{C_p}\prod_{i=1}^{2p-1}(x-h_{1,i})^2\prod_{i=1}^{2p-1}(x-h_{2,i}) \qquad (\widetilde{C_p}\neq 0).$$

(Here x corresponds to $[\omega]$ and y to $[H]$.)

The triplet vertex algebra $\mathcal{W}_{p,2}$

Logarithmic extension of $(p,2)$–minimal models.

The triplet vertex algebra $\mathcal{W}_{p,2}$ is $\mathcal{W}$–algebra of type

$$\mathcal{W}(2,h,h,h), \quad (h=(2n+1)(pn+p-1)).$$

Theorem (D.A, A.M)

(i) *Every Virasoro minimal model for central charge $c_{p,2}$ is a module for $\mathcal{W}_{p,2}$.*

(ii) *$\mathcal{W}_{p,2}$ has exactly $4p+\frac{p-1}{2}$ irreducible modules.*

(iii) *$\mathcal{W}_{p,2}$ is C_2–cofinite.*

(iv) *$\mathcal{W}_{p,2}$ is irrational and admits logarithmic modules.*

The triplet vertex algebra $\mathcal{W}_{p,2}$

Logarithmic extension of $(p,2)$–minimal models.

The triplet vertex algebra $\mathcal{W}_{p,2}$ is $\mathcal{W}$–algebra of type

$$\mathcal{W}(2,h,h,h), \quad (h=(2n+1)(pn+p-1)).$$

Theorem (D.A, A.M)

(i) *Every Virasoro minimal model for central charge $c_{p,2}$ is a module for $\mathcal{W}_{p,2}$.*

(ii) *$\mathcal{W}_{p,2}$ has exactly $4p+\frac{p-1}{2}$ irreducible modules.*

(iii) *$\mathcal{W}_{p,2}$ is C_2–cofinite.*

(iv) *$\mathcal{W}_{p,2}$ is irrational and admits logarithmic modules.*

The Zhu's algebra $A(\mathcal{W}_{3,2})$

Let $p = 3$. Then $\mathcal{W}_{3,2}$ is called the triplet $c = 0$ vertex algebra.
Generators of $A(\mathcal{W}_{3,2})$: $[\omega], [H], [E], [F]$.

Theorem

Zhu's algebra $A(\mathcal{W}_{3,2})$ decomposes as a direct sum:

$$A(\mathcal{W}_{3,2}) = \bigoplus_{h \in S^{(2)}} \mathbb{M}_h \oplus \bigoplus_{h \in S^{(1)}} \mathbb{I}_h \oplus \mathbb{C}_{-1/24},$$

$$S^{(2)} = \{5, 7, \tfrac{10}{3}, \tfrac{33}{8}, \tfrac{21}{8}, \tfrac{35}{24}\}, S^{(1)} = \{0, 1, 2, \tfrac{1}{3}, \tfrac{1}{8}, \tfrac{5}{8}, \tfrac{-1}{24}\}.$$

$\mathbb{M}_h$ is ideal isomorphic to $M_2(\mathbb{C})$, $h \in S^{(2)}$,

$\mathbb{I}_h$ is 2–dimensional ideal, $h \in S^{(1)}$, $h \neq 0$, $h \neq -1/24$,

$\mathbb{C}_{-1/24}$ is 1-dimensional ideal,

$\mathbb{I}_0$ is 3–dimensional ideal.

The triplet vertex algebra $\mathcal{W}_{p,p'}$

Conjecture

Assume that $(p,p')=1$. The vertex algebra $\mathcal{W}_{p,p'}$ is C_2–cofinite with $2pp'+\frac{(p-1)(p'-1)}{2}$–irreducible modules.

$N = 1$ Neveu-Schwarz vertex operator superalgebras

The $N = 1$ Neveu-Schwarz algebra is the Lie superalgebra

$$\mathbf{ns} = \bigoplus_{n\in\mathbb{Z}} \mathbb{C}L(n) \bigoplus \bigoplus_{m\in\frac{1}{2}+\mathbb{Z}} \mathbb{C}G(m) \bigoplus \mathbb{C}C$$

with commutation relations ($m, n \in \mathbb{Z}$):

$$[L(m), L(n)] = (m-n)L(m+n) + \delta_{m+n,0}\frac{m^3-m}{12}C,$$
$$[G(m+\frac{1}{2}), L(n)] = (m+\frac{1}{2}-\frac{n}{2})G(m+n+\frac{1}{2}),$$
$$\{G(m+\frac{1}{2}), G(n-\frac{1}{2})\} = 2L(m+n) + \frac{1}{3}m(m+1)\delta_{m+n,0}C,$$
$$[L(m), C] = 0, \quad [G(m+\frac{1}{2}), C] = 0.$$

$N = 1$ Neveu-Schwarz vertex operator superalgebras

We say that the vertex operator superalgebra V is a $N = 1$ Neveu-Schwarz vertex operator superalgebra if:

There exists $\tau \in V_{3/2}$ (superconformal vector) such that

$$Y(\tau, z) = \sum_{n \in \mathbb{Z}+1/2} G(n) z^{-n-3/2}, \quad G(n) \in \mathrm{End}(V)$$

$$Y(\omega, z) = \sum_{n \in \mathbb{Z}} L(n) z^{-n-2}, \; \omega = 1/2 G(-1/2)\tau, \quad L(n) \in \mathrm{End}(V)$$

and $G(n)$, $L(n)$ satisfy the commutation relations for $N = 1$ Neveu-Schwarz algebra.

$N = 1$ Neveu-Schwarz vertex operator superalgebras

The simplest examples of $N = 1$ vertex operator superalgebras are **ns**-modules $L^{\mathbf{ns}}(c, 0)$, $c \neq 0$,

where for any $(c, h) \in \mathbb{C}^2$ we denote by $L^{\mathbf{ns}}(c, h)$ the corresponding irreducible highest weight **ns**–module with central charge c and highest weight h.

The vertex operator superalgebra $L^{\mathbf{ns}}(c, 0)$, $c \neq 0$ is simple.

Set

$$c_{p,q} = \frac{3}{2}(1 - \frac{2(p-q)^2}{pq}),$$

$$h^{r,s}_{p,q} = \frac{(sp - rq)^2 - (p-q)^2}{8pq}.$$

$N = 1$ Neveu-Schwarz vertex operator superalgebras

The simplest examples of $N = 1$ vertex operator superalgebras are **ns**-modules $L^{\mathbf{ns}}(c, 0)$, $c \neq 0$,

where for any $(c, h) \in \mathbb{C}^2$ we denote by $L^{\mathbf{ns}}(c, h)$ the corresponding irreducible highest weight **ns**–module with central charge c and highest weight h.

The vertex operator superalgebra $L^{\mathbf{ns}}(c, 0)$, $c \neq 0$ is simple.

Set

$$c_{p,q} = \frac{3}{2}(1 - \frac{2(p-q)^2}{pq}),$$

$$h^{r,s}_{p,q} = \frac{(sp - rq)^2 - (p-q)^2}{8pq}.$$

Fermionic vertex operator superalgebra $\mathcal{F}$

We consider the Clifford algebra CL, generated by
$\{\phi(n), n \in \frac{1}{2} + \mathbb{Z}\} \cup \{1\}$
and relations

$$\{\phi(n), \phi(m)\} = \delta_{n,-m}, \quad n, m \in \tfrac{1}{2} + \mathbb{Z}.$$

Let $\mathcal{F}$ be the CL–module generated by the vector $\mathbf{1}$ such that

$$\phi(n)\mathbf{1} = 0, \ \ n > 0.$$

The field

$$Y(\phi(-\tfrac{1}{2})\mathbf{1}, z) = \phi(z) = \sum_{n \in \frac{1}{2}+\mathbb{Z}} \phi(n) z^{-n-\frac{1}{2}},$$

generates the unique vertex operator superalgebra structure on $\mathcal{F}$.

Vertex operator superalgebra $V_L \otimes \mathcal{F}$

Let $V_L = M(1) \otimes \mathbb{C}[L]$ be the vertex superalgebra associated to the lattice

$$L = \mathbb{Z}\alpha, \quad \langle \alpha, \alpha \rangle = 2m + 1, \ m \in \mathbf{Z}_{>0}.$$

We shall consider the vertex superalgebras

$$V_L \otimes \mathcal{F}, \quad SM(1) = M(1) \otimes \mathcal{F}$$

Vertex operator superalgebra $V_L \otimes \mathcal{F}$

$SM(1)$ and $V_L \otimes \mathcal{F}$ are $N = 1$ Neveu-Schwarz vertex operator superalgebras of central charge $c_{2m+1,1}$.

Superconformal vector

$$\tau = \frac{1}{\sqrt{2m+1}}(\alpha(-1)\mathbf{1} \otimes \phi(-\tfrac{1}{2})\mathbf{1} + 2m\mathbf{1} \otimes \phi(-\tfrac{3}{2})\mathbf{1}).$$

$$\omega = \frac{1}{2}\tau_0\tau.$$

Screening operators

$$Q = Res_z Y(e^{\alpha} \otimes \phi(-1/2)\mathbf{1}, z),$$

$$\widetilde{Q} = Res_z Y(e^{-\alpha/2m+1} \otimes \phi(-1/2)\mathbf{1}, z).$$

Super-triplet vertex algebra $\mathcal{SW}(m)$

Define:

$$\mathcal{SW}(m) = \mathrm{Ker}_{V_L \otimes \mathcal{F}} \widetilde{Q}, \quad \overline{SM(1)}_m = \mathrm{Ker}_{SM(1)} \widetilde{Q}.$$

Generators of $\mathcal{SW}(m)$:

$$\tau, F = e^{-\alpha} \otimes \phi(-1/2), \ H = QF, \ E = Q^2 F.$$

$\mathcal{SW}(m)$ is super $\mathcal{W}$–algebra of type $\mathcal{W}(\frac{3}{2}, 2m + \frac{1}{2}, 2m + \frac{1}{2}, 2m + \frac{1}{2})$.

$\overline{SM(1)}_m$ is generated by τ and H.

Super-triplet vertex algebra $\mathcal{SW}(m)$

Define:

$$\mathcal{SW}(m) = \mathrm{Ker}_{V_L \otimes \mathcal{F}} \widetilde{Q}, \quad \overline{SM(1)}_m = \mathrm{Ker}_{SM(1)} \widetilde{Q}.$$

Generators of $\mathcal{SW}(m)$:

$$\tau, F = e^{-\alpha} \otimes \phi(-1/2), \ H = QF, \ E = Q^2 F.$$

$\mathcal{SW}(m)$ is super $\mathcal{W}$–algebra of type $\mathcal{W}(\frac{3}{2}, 2m + \frac{1}{2}, 2m + \frac{1}{2}, 2m + \frac{1}{2})$.
$\overline{SM(1)}_m$ is generated by τ and H.

Zhu's algebra $A(\overline{SM(1)}_m)$

Theorem

Zhu's algebra $A(\overline{SM(1)}_m)$ is isomorphic to the commutative algebra $\mathbb{C}[x,y]/\langle Q(x,y)\rangle$, where $\langle Q(x,y)\rangle$ is the principal ideal generated by

$$Q(x,y) = y^2 - D_m \prod_{i=0}^{2m} (x - h^{2i+1,1}), \quad (D_m \neq 0).$$

(Here x and y correspond to $[\omega]$ and $[G(-1/2)H]$.)

Super-triplet vertex algebra $\mathcal{SW}(m)$

For a vertex operator superalgebra V let σ be the canonical $\mathbb{Z}_2$–automorphism, and $A_\sigma(V)$ twisted Zhu's algebra.

Theorem (D.A; A. M)

Assume that $m \geq 1$. Then

(i) *The vertex operator superalgebra $\mathcal{SW}(m)$ is C_2–cofinite.*

(ii) *$\mathcal{SW}(m)$ is irrational and admits logarithmic modules.*

(iii) *$\mathcal{SW}(m)$ has $2m+1$ irreducible modules.*

(iv) *$\mathcal{SW}(m)$ has $2m+1$ $\mathbb{Z}_2$–graded irreducible σ–twisted modules.*

(v) $\dim \mathcal{P}(\mathcal{SW}(m)) = \dim A_\sigma(\mathcal{SW}(m))$.

Super-triplet vertex algebra $\mathcal{SW}(m)$

For a vertex operator superalgebra V let σ be the canonical $\mathbb{Z}_2$–automorphism, and $A_\sigma(V)$ twisted Zhu's algebra.

Theorem (D.A; A. M)

Assume that $m \geq 1$. Then

(i) *The vertex operator superalgebra $\mathcal{SW}(m)$ is C_2–cofinite.*

(ii) *$\mathcal{SW}(m)$ is irrational and admits logarithmic modules.*

(iii) *$\mathcal{SW}(m)$ has $2m+1$ irreducible modules.*

(iv) *$\mathcal{SW}(m)$ has $2m+1$ $\mathbb{Z}_2$–graded irreducible σ–twisted modules.*

(v) $\dim \mathcal{P}(\mathcal{SW}(m)) = \dim A_\sigma(\mathcal{SW}(m))$.

Super-triplet vertex algebra $\mathcal{SW}(m)$

For a vertex operator superalgebra V let σ be the canonical $\mathbb{Z}_2$–automorphism, and $A_\sigma(V)$ twisted Zhu's algebra.

Theorem (D.A; A. M)

Assume that $m \geq 1$. Then

(i) *The vertex operator superalgebra $\mathcal{SW}(m)$ is C_2–cofinite.*

(ii) *$\mathcal{SW}(m)$ is irrational and admits logarithmic modules.*

(iii) *$\mathcal{SW}(m)$ has $2m+1$ irreducible modules.*

(iv) *$\mathcal{SW}(m)$ has $2m+1$ $\mathbb{Z}_2$–graded irreducible σ–twisted modules.*

(v) $\dim \mathcal{P}(\mathcal{SW}(m)) = \dim A_\sigma(\mathcal{SW}(m))$.

Super-triplet vertex algebra $\mathcal{SW}(m)$

For a vertex operator superalgebra V let σ be the canonical $\mathbb{Z}_2$–automorphism, and $A_\sigma(V)$ twisted Zhu's algebra.

Theorem (D.A; A. M)

Assume that $m \geq 1$. Then

(i) *The vertex operator superalgebra $\mathcal{SW}(m)$ is C_2–cofinite.*

(ii) *$\mathcal{SW}(m)$ is irrational and admits logarithmic modules.*

(iii) *$\mathcal{SW}(m)$ has $2m+1$ irreducible modules.*

(iv) *$\mathcal{SW}(m)$ has $2m+1$ $\mathbb{Z}_2$–graded irreducible σ–twisted modules.*

(v) $\dim \mathcal{P}(\mathcal{SW}(m)) = \dim A_\sigma(\mathcal{SW}(m))$.

Super-triplet vertex algebra $\mathcal{SW}(m)$

For a vertex operator superalgebra V let σ be the canonical $\mathbb{Z}_2$–automorphism, and $A_\sigma(V)$ twisted Zhu's algebra.

Theorem (D.A; A. M)

Assume that $m \geq 1$. Then

(i) *The vertex operator superalgebra $\mathcal{SW}(m)$ is C_2–cofinite.*

(ii) *$\mathcal{SW}(m)$ is irrational and admits logarithmic modules.*

(iii) *$\mathcal{SW}(m)$ has $2m+1$ irreducible modules.*

(iv) *$\mathcal{SW}(m)$ has $2m+1$ $\mathbb{Z}_2$–graded irreducible σ–twisted modules.*

(v) $\dim \mathcal{P}(\mathcal{SW}(m)) = \dim A_\sigma(\mathcal{SW}(m))$.

Super-triplet vertex algebra $\mathcal{SW}(m)$

For a vertex operator superalgebra V let σ be the canonical $\mathbb{Z}_2$–automorphism, and $A_\sigma(V)$ twisted Zhu's algebra.

Theorem (D.A; A. M)

Assume that $m \geq 1$. Then

(i) *The vertex operator superalgebra $\mathcal{SW}(m)$ is C_2–cofinite.*

(ii) *$\mathcal{SW}(m)$ is irrational and admits logarithmic modules.*

(iii) *$\mathcal{SW}(m)$ has $2m+1$ irreducible modules.*

(iv) *$\mathcal{SW}(m)$ has $2m+1$ $\mathbb{Z}_2$–graded irreducible σ–twisted modules.*

(v) $\dim \mathcal{P}(\mathcal{SW}(m)) = \dim A_\sigma(\mathcal{SW}(m))$.

Zhu's algebra $A(\mathcal{SW}(m))$

Theorem

Zhu's algebra decomposes as a sum of ideals

$$A(\mathcal{SW}(m)) = \bigoplus_{i=2m+1}^{3m} \mathbb{M}_{h^{2i+1,1}} \oplus \bigoplus_{i=0}^{m-1} \mathbb{I}_{h^{2i+1,1}} \oplus \mathbb{C}_{h^{2m+1,1}},$$

where $\mathbb{M}_{h^{2i+1,1}} \cong M_2(\mathbb{C})$, $\dim(\mathbb{I}_{h^{2i+1,1}}) = 2$ *and* $\mathbb{C}_{h^{2m+1,1}}$ *is one-dimensional.*

Logarithmic modules

Corollary

For every $0 \leq i \leq m-1$, there exits the logarithmic, self-dual, $\mathbb{Z}_{\geq 0}$–graded $\mathcal{SW}(m)$–module $\mathcal{SP}_i^+$ such that the top component $\mathcal{SP}_i^+(0)$ is two-dimensional and $L(0)$ acts on $\mathcal{SP}_i^+(0)$ (in some basis) as

$$\begin{pmatrix} h^{2i+1,1} & 1 \\ 0 & h^{2i+1,1} \end{pmatrix}.$$

Thank you!